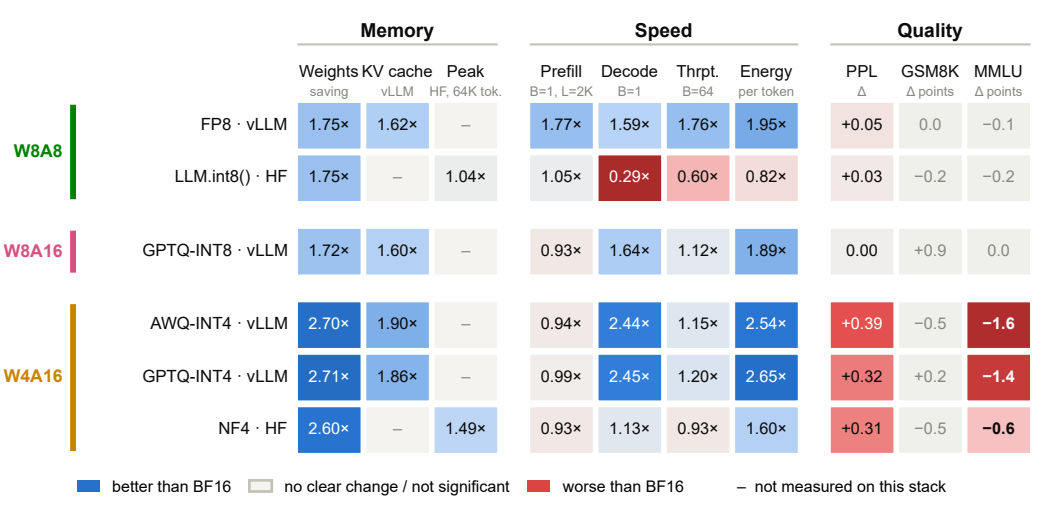
Memory
Speed
Quality
Weights saving
KV cache vLLM
Peak HF, 64K tok.
Prefill B=1, L=2K
Decode B=1
Thrpt. B=64
Energy per token
PPL Δ
GSM8K Δ points
MMLU Δ points
W8A8
FP8 · vLLM
1.75× 1.62× – 1.77× 1.59× 1.76× 1.95× +0.05 0.0 −0.1
LLM.int8() · HF
1.75× – 1.04× 1.05× 0.29× 0.60× 0.82× +0.03 −0.2 −0.2
W8A16
GPTQ-INT8 · vLLM
1.72× 1.60× – 0.93× 1.64× 1.12× 1.89× 0.00 +0.9 0.0
W4A16
AWQ-INT4 · vLLM
2.70× 1.90× – 0.94× 2.44× 1.15× 2.54× +0.39 −0.5 −1.6
GPTQ-INT4 · vLLM
2.71× 1.86× – 0.99× 2.45× 1.20× 2.65× +0.32 +0.2 −1.4
NF4 · HF
2.60× – 1.49× 0.93× 1.13× 0.93× 1.60× +0.31 −0.5 −0.6
better than BF16
no clear change / not significant
worse than BF16
– not measured on this stack